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Math Cell

Syntax

Assignment including number with units:

$$x = 10 [m]$$

Multiple assignments:

$$y = 2 [m], z = 5 [N]$$

Query to evaluate expression:

$$z \cdot (x \cdot y)^{-1} = 0.25 [Pa]$$

Specifying preferred units for a query result:

$$z \cdot (x \cdot y)^{-1} = [psi] = 3.63 \times 10^{-5} [psi]$$

Combined assignment and query:

$$Pressure = z \cdot (x \cdot y)^{-1} = 0.25 [Pa]$$

Predefined variables (cannot be redefined, type **pi** for π):

$$\pi = 3.14159265358979 \quad e = 2.71828182845905 \quad i = \sqrt{-1}$$

Comment mode toggled with the quotation mark key:

$$\sigma = \frac{M \cdot y}{I} \quad \text{Bending stress equation}$$

The expression above can be called as a function:

$$\sigma (M = 1 [kN \cdot m], y = 1 [mm], I = 1 [m^4]) = 1 [Pa]$$

Math Field Keyboard Shortcuts

$\wedge$	Add exponent
$_$	Add subscript
$*$	Multiply
$/$	Divide
Ctrl/Cmd + E	Insert scientific notation
Tab	Move out of exponent or subscript
"	Toggle comment mode
Type sqrt	Insert square root
Spell out a Greek character to insert (may be preceded by \ to insert immediately): alpha, beta, gamma, delta, epsilon, zeta, eta, theta, iota, kappa, lambda, mu, nu, xi, pi, rho, sigma, tau, phi, chi, psi, omega, Gamma, Delta, Theta, Lambda, Xi, Pi, Sigma, Upsilon, Phi, Psi, and Omega	

Type \int	Insert integral
Type \der	Insert derivative
Type \dder	Insert double derivative
Type \tder	Insert triple derivative
Type \sum	Insert summation notation
Type \prod	Insert product notation
Type \hat	Add hat accent to variable name
Type \bar	Add bar accent to variable name
Type \vec	Add vector accent to variable name
Type \dot	Add dot accent to variable name
Type \ddot	Add double dot accent to variable name
'	Add prime to variable name

← or → or ↓ or ↑ Navigate within math field

[Live Tutorial](#)

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Documentation Cell

Normal B I U A [color] [background color] [list] [link] [math] [table]

Summary

A documentation cell is a rich text editor that can contain images, links, and mathematical expressions.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

- Set heading level
 - Toggle bold text (Ctrl/Cmd + B)
 - Toggle italics text (Ctrl/Cmd + I)
 - Toggle underline text (Ctrl/Cmd + U)
 - Set text color
 - Set text background color
 - Insert numbered list
 - Insert bulleted list
 - Insert link (Ctrl/Cmd + K)
 - Insert image (images may also be pasted)
 - Insert formula (uses LaTeX notation) (Ctrl/Cmd + E)
 - Remove text formatting
- Ctrl/Cmd + Z Undo edit in current field
Ctrl + Y or Cmd + Shift + Z Redo edit in current field

Piecewise Expression Cell

$$\text{triangle} = \begin{cases} x + 1 & \text{if } -1 \leq x \leq 0 \\ -(x - 1) & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

$\text{triangle}(x = -0.5) = 0.5$

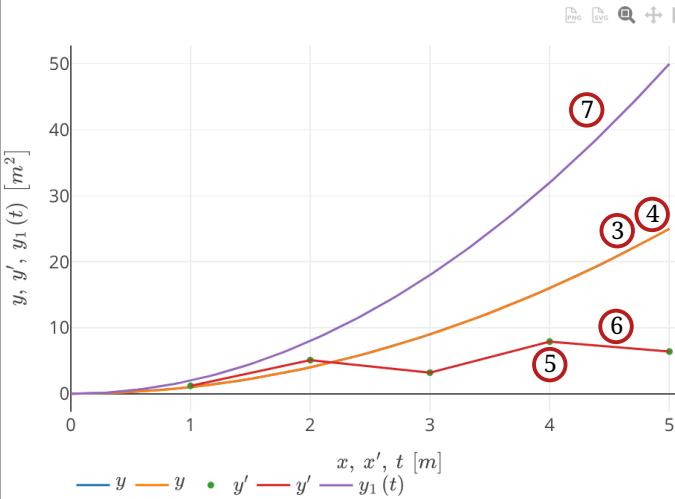
- Expression is assigned to this variable
- Values that are assigned to the variable
- Conditions (the first true condition is used)
- Value used if no condition is true
- The variable can be used like any other expression

< then = inserts ≤ > then = inserts ≥

Plot Cell

$$y = x^2 \quad 1$$

$$x' = [1 \ 2 \ 3 \ 4 \ 5] \cdot 1 [m], y' = [1.2 \ 5.1 \ 3.2 \ 7.9 \ 6.4] \cdot 1 [m^2] \quad 2$$



log x log y 1:1 Ratio Copy Data

$$y(0 [m] \leq x \leq 5 [m]) = \quad 3$$

$$y(0 [m] \leq x \leq 5 [m]) \text{ with 200 points} = \quad 4$$

$$(x', y') = \quad 5$$

$$(x', y') \text{ as lines} = \quad 6$$

$$(t, 2 \cdot t^2) \text{ for } (0 [m] \leq t \leq 5 [m]) = \quad 7$$

- 1 Function definition
- 2 Scatter points definition
- 3 Function plot
- 4 Function plot w/ num of points specified (default 51)
- 5 Scatter plot as points (default)
- 6 Scatter plot as lines
- 7 Parametric plot

< then = inserts ≤ > then = inserts ≥

[Plotting Live Tutorial](#)

Global Keyboard Shortcuts

Ctrl/Cmd + S	Save current sheet to file
Ctrl/Cmd + Shift + S	Save as .pro
Ctrl/Cmd + O	Open file
Ctrl/Cmd + N	Open new tab .pro
Ctrl/Cmd + W	Close current tab .pro
Ctrl/Cmd + Tab	Switch to next tab .pro
Ctrl/Cmd + Shift + Tab	Switch to previous tab .pro
Ctrl/Cmd + ↑	Select cell above current cell
Ctrl/Cmd + ↓	Select cell below current cell
Ctrl/Cmd + D	Delete current cell
Shift + Enter	Insert math cell below current cell
Ctrl/Cmd + Enter	Insert custom cell below current cell
Esc	Unselect all cells

System Solve Cell

Symbolic System Solve

$$\begin{aligned} \text{System} = & \begin{cases} x^2 - 5 \cdot x + 6 = 0 \\ y + x + c = 10 \end{cases} \quad 1 \\ & + \\ \text{Solve for: } & x, y \quad 2 \end{aligned}$$

$$\begin{aligned} \text{Solution} = & \begin{cases} x = 2 & 3 \\ y = 8 - c & 7 - c \end{cases} \\ \text{Selected Solution: } & \bullet \quad 3 \end{aligned}$$

$$y = 8 - c \quad 6$$

Numerical System Solve

$$\begin{aligned} \text{System} = & \begin{cases} H = 10 [m] \\ L = 15 [m] \\ H = 2 \cdot a \cdot \ln\left(\frac{(h+a)}{a}\right) + \sqrt{\left(\frac{(h+a)}{a}\right)^2 - 1} \\ L = 2 \cdot a \cdot \sinh\left(\frac{H}{2 \cdot a}\right) \end{cases} \quad 1 \\ & + \\ \text{Solve for: } & h \approx 3 [m], a \approx 2 [m], H \approx 10 [m], L \approx 15 [m] \quad 2 \quad 4 \end{aligned}$$

$$h = 5.02633261648315 [m] \quad 6$$

- 1 Specify system of equations
- 2 Specify variables to solve for
- 3 If more than one solution, choose desired solution
- 4 Initial guesses to trigger numerical solve (~ inserts ≈)
- 5 Define all variables needed by numerical solve
- 6 Results are available in the rest of the sheet

[Equation Solving Live Tutorial](#)

Selector Table Cell



		Area	l	r	
			[m]	[m]	
		l^2	2		
1	Circle	π · r^2		1.5	

$$\text{Area} = 7.06858347057703 [m^2] \quad 6$$

- 1 Selected row inserts variable values into rest of sheet
- 2 Each column sets the value for a variable
- 3 First row sets variable names
- 4 Units can optionally be specified in the units row
- 5 Row specific documentation can be added using Ⓢ
- 6 Column variables are available in the rest of the sheet

Data Table Cell

Import Spreadsheet Export CSV Copy Data Add Interpolation Add Polyfit

l	w	$area = l \cdot w$
$[m]$	$[m]$	$[m^2]$
1.5	10	15
2	12	24
3	11	33

$l^T = [1.5 [m] \ 2 [m] \ 3 [m]]$
 $area^T = [15 [m^2] \ 24 [m^2] \ 33 [m^2]]$

- Input columns assign column contents to a variable
- Output columns perform row-by-row calculations
- Optional units may be specified for input columns
- Input/output column variables become vectors
- Data can be imported from files (CSV or XLSX)
- Data can be exported as CSV files
- Input columns can be interpolated
- Polynomials can be fit to input columns

[Data Table Live Tutorial](#)

Fluid Properties Cell

Fluid: Ethanol Use sheet fluid
 Output: T - Temperature - [K]
 Input 1: P - Pressure - [Pa]
 Input 2: Q - Molar vapor quality - [mol/mol]
 Function Name: EthanolTGivenPQ Use fluid name in function name

$EthanolTGivenPQ(1 [bar], 0) = 78.0871426243792 [degC]$

- Select fluid
- Select output
- Select the 0, 2, or 3 inputs (0 for const., 3 for humid air)
- Function or constant name used in the sheet
- Function is called similarly to a built-in function
- Use toggle to lock all fluid cells to the same fluid

[Fluid Properties Live Tutorial](#)

Insert Sheet Command

Click the insert sheet button to insert an existing sheet

Select the "Link Sheet" option to link to one or more external sheets **.pro**

Linked sheets are linked by relative file path and link will be preserved as long as files are kept together **.pro**

Code Cell

Scalar Example

$atan2([mm], [mm]) = [rad]$

☐ Use SymPy Mode

```

1 from math import atan2
2 def calculate(y, x):
3     # y and x will come in with mm units
4     print(f"{x=}")
5     print(f"{y=}")
6     # Return value assumed to be in radians
7     return atan2(y, x)

```

$x=1000.0$
 $y=1732.0508075688772$

$atan2(\sqrt{3} \cdot 1 [m], 1 [m]) = [deg] = 60 [deg]$

- Name of function that will be created
- Units for inputs (will be converted before function call)
- Units for output value within the function
- Function can be called like built in functions
- Function code, function must be named **calculate**
- Click info button to see what packages are available
- SymPy mode treats inputs and output as SymPy objects
- Print function output appears below code

Matrix Example

$prod([any]) = [none]$

☐ Use SymPy Mode

```

1 import numpy as np
2 def calculate(value):
3     # Matrices come in as 2D NumPy arrays
4     print(f"{value=}")
5     return np.prod(value)

```

$value=array([[1., 2., 3.]])$

$prod([1 \ 2 \ 3]) = 6$

- Use [any] for inputs that can have any units
- [none] indicates input or output will be unitless

Custom Units Example

$prod([any]) = [any]$

☐ Use SymPy Mode

```

1 import numpy as np
2 import sympy as sp
3 def calculate(value):
4     return np.prod(value)
5
6 # If output dims are [any]
7 # A custom_dims function is required
8 def custom_dims(value):
9     return sp.Mul(*value)

```

$prod([1 [m] \ 2 [m] \ 3 [m]]) = 6 [m^3]$

- [any] for output units requires a **custom_dims** function
- custom_dims** inputs are always SymPy objects

[Code Cell Live Tutorial](#)

f(x)

Built-in Functions

General

$\sqrt{x}$	Square root, type sqrt to insert
x^y	Exponent, press  to insert
$\ln(x)$	Natural logarithm (base e)
$\log(x)$	Base 10 logarithm
$\log_b(x)$	Base b logarithm, type log then press 
$ x $	Absolute value
$\text{round}(x)$	Round to nearest integer
$\text{ceil}(x)$	Round up to nearest integer
$\text{floor}(x)$	Round down to nearest integer
$n!$	Factorial

Trigonometry

$\sin(x)$	Sine, input unitless or angle
$\cos(x)$	Cosine, input unitless or angle
$\tan(x)$	Tangent, input unitless or angle
$\sec(x)$	Secant, input unitless or angle
$\cos(x)$	Cosecant, input unitless or angle
$\cot(x)$	Cotangent, input unitless or angle
$\arcsin(x)$	Inverse sine, input unitless, output angle
$\arccos(x)$	Inverse cosine, input unitless, output angle
$\arctan(x)$	Inverse tangent, input unitless, output angle
$\sinh(x)$	Hyperbolic sine, unitless input/output
$\cosh(x)$	Hyperbolic cosine, unitless input/output
$\tanh(x)$	Hyperbolic tangent, unitless input/output

Complex Numbers

$\text{real}(x)$	Real part of a complex number
$\text{imag}(x)$	Imaginary part of a complex number
$\text{angle}(x)$	Angle of a complex number
$\text{conj}(x)$	Complex conjugate of a complex number

Matrices and Vectors

$\text{numrows}(x)$ or $\text{numcols}(x)$	Number of rows or columns
$\text{min}(x)$ or $\text{max}(x)$	Min or max of list or vector
$\text{count}(x)$	Number of entries in a matrix/vector
$\text{sum}(x)$	Sum of values of list or matrix/vector
$\text{average}(x)$	Average of values of list or matrix/vector
$\text{stdev}(x)$ or $\text{stdevp}(x)$	Sample or population std. dev.
$\text{range}(n)$	Return vector of integers 1 to n
$\text{range}(m, n)$	Return vector of integers m to n
$\text{range}(m, n, s)$	Return vector of integers m to n , step s
$\ A\ $ or $\text{norm}(A)$	Euclidean norm
$\det(A)$	Determinant
$\text{dot}(v_1, v_2)$	Dot product
$v_1 \times v_2$	Cross product for 3 by 1 or 1 by 3, type 
A^T	Matrix transpose  + 
A^{-1}	Matrix inverse

$A_{3,4}$ or $A_{1:\text{end},4:2:12}$ Matrix indexing examples

[Matrices and Vectors Live Tutorial](#) 

Calculus See function toolbar, f(x), for calculus operators

Built-in Units

Units can be used in their long and short forms. Most units support long prefixes in their long form and short prefixes in their short form (e.g., kilogram or kg). Long form units generally support their singular and plural forms (e.g., meter or meters). Units may be combined as needed using exponents, multiplication, and division, as in: $\left[\frac{\text{kg}}{\text{m}^3}\right]$ or $[\text{kg} \cdot \text{m}^{-3}]$

Supported Units

Base	Unit
Length	meter (m), inch (in), foot (ft), yard (yd), mile (mi), link (li), rod (rd), chain (ch), angstrom, mil
Surface area	m2, sqin, sqft, sqyd, sqmi, sqrd, sqch, sqmil, acre, hectare
Volume	m3, litre (l, L, lt, liter), cc, cuin, cuft, cuyd, teaspoon, tablespoon
Liquid vol.	minim, fluid dram (fldr), fluid ounce (floz), gill (gi), cup (cp), pint (pt), quart (qt), gallon (gal), beer barrel (bbl), oil barrel (obl), hogshead, drop (gtt)
Angles	rad (radian), deg (degree), grad (gradian), cycle, arcsec (arcsecond), arcmin (arcminute)
Time	second (s, secs, seconds), minute (min, mins, minutes), hour (h, hr, hrs, hours), day (days), week (weeks), month (months), s year (years), decade (decades), century (centuries), millennium (millennia)
Frequency	hertz (Hz)
Mass	gram(g), tonne, ton, grain (gr), dram (dr), ounce (oz), pound mass (lbm, lb, lbs), hundredweight (cwt), stick, stone
Elec. current	ampere (A)
Temperature	kelvin (K), celsius (degC), fahrenheit (degF), rankine (degR)
Amount sub.	mole (mol)
Luminous int.	candela (cd)
Force	newton (N), dyne (dyn), pound force (lbf), kip, kilogram force (kgf), gram force (gf)
Energy	joule (J), erg, Wh, BTU (Btu), electronvolt (eV)
Power	watt (W), hp, tonrefrigeration (TR, RT, TOR, toncooling), boiler hp (BHP), MBH, MMBH
Pressure	Pa, psi, atm, torr, bar, mmHg, mmH2O, cmH2O, inHg, inH2O, ftH2O, kpsi (ksi), Mpsi
Elec. / Mag.	ampere (A), coulomb (C), watt (W), volt (V), ohm, farad (F), weber (Wb), tesla (T), henry (H), siemens (S), electronvolt (eV)
Viscosity	poise (P), reyn, ureyn, stokes (St)
Binary	bits (b), bytes (B)

Supported Unit Prefixes

Name	Abbr.	Value	Name	Abbr.	Value
deca	da	1e1	deci	d	1e-1
hecto	h	1e2	centi	c	1e-2
kilo	k	1e3	milli	m	1e-3
mega	M	1e6	micro	u	1e-6
giga	G	1e9	nano	n	1e-9
tera	T	1e12	pico	p	1e-12
peta	P	1e15	femto	f	1e-15
exa	E	1e18	atto	a	1e-18
zetta	Z	1e21	zepto	z	1e-21
yotta	Y	1e24	yocto	y	1e-24